

TESTING THE HOMOGENEITY OF VARIANCES AGAINST AN ORDERED ALTERNATIVE

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SUMMARY

One of the assumptions in the general normal linear regression model is the homogeneity of variances of the disturbances. Often a realistic alternative to homogeneity of variances is a set of ordered variances and accordingly we consider the following general problem.

Independent samples, not necessarily of the same size, from k normal populations are given and we wish to test the homogeneity of the k variances against an ordered alternative.

Vincent (1961) considered this problem and for equal sample sizes derived a test which for a special choice of the alternative can 'approximately' be regarded as the uniformly most powerful unbiased test.

In this paper we derive the most powerful invariant test for this problem. For Vincent's special choice of the alternative this test is the uniformly most powerful invariant test, and for equal sample sizes this test is also identical with Vincent's test.

This raises the question of why Vincent's test is only 'approximately' the uniformly most powerful unbiased test and we show that this qualification is indeed superfluous. Furthermore Vincent's test can also be extended to unequal sample sizes.

The limiting distribution of the proposed test statistic is normal and for small sample sizes of equal size, we give an approximation based on the Beta distribution which will usually suffice in practice.

Finally the proposed test is compared asymptotically with Bartlett's test for the homogeneity of variances.

1. Most Powerful Invariant Test for Homoscedasticity

Let $X_{11}, \dots, X_{1n_1}, \dots, X_{k1}, \dots, X_{kn_k}$ be independent samples from k normal populations with unknown means and variances $\sigma_1^2, \dots, \sigma_k^2$. We wish to test the null hypothesis

$$H_0: \sigma_1^2 = \dots = \sigma_k^2$$

against an ordered alternative, say

$$H_a: \sigma_1^2 \leq \dots \leq \sigma_k^2$$

with at least one strict inequality.

The statistics $\bar{X}_1, \dots, \bar{X}_k; S_1^2, \dots, S_k^2$ with

$$\bar{X}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} X_{ij}$$

$$S_i^2 = \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2, \quad i=1, \dots, k,$$

are sufficient statistics for the parameters $\mu_1, \dots, \mu_k; \sigma_1^2, \dots, \sigma_k^2$ respectively. The problem under consideration remains invariant under location and scale transformations and for these transformations the above sufficient statistics have as maximal invariants $F_2, \dots, F_k$, with $F_i = S_i^2 / S_1^2$ (see Lehmann (1959), page 216). Any test depending on the X_{ij} only through these maximal invariants is known to be an invariant test. Considering only these maximal invariants and using the Neyman-Pearson criterion, the most powerful invariant critical region for testing H_0 against H_a is readily found to be of the form

$$(1 + \sum_{i=2}^k F_i) / (1 + \sigma_1^2 \sum_{i=2}^k F_i / \sigma_i^2) \geq c, \text{ where } c \text{ is constant.}$$

Expressing H_a in the form

$$H_a: \sigma_i^2 = \sigma_1^2 [1 - a_i \theta]^{-1}, \quad \theta > 0, \quad a_1 = 0, \quad 0 \leq a_i < \frac{1}{\theta},$$

a_i nondecreasing in i with at least one $a_i > 0$, this critical region reduces to

$$B = \sum_{i=2}^k a_i S_i^2 / \sum_{i=1}^k S_i^2 \geq c.$$

Since B is independent of θ , the corresponding test is the uniformly most powerful invariant test for testing H_0 against H_a when the a_i are known.

The alternative H_a can be put in the form

$$\frac{1}{\sigma_1^2} - \frac{1}{\sigma_i^2} = \frac{\theta a_i}{\sigma_1^2}$$

which means that the distances between precisions are known, apart from a common factor θ/σ_1^2 . The case of constant differences between precisions is given by $a_i = i-1$ and this is the alternative considered by Vincent (1961).

Especially in time series the alternative to homogeneity of variances may be one

of proportionate growth, that is

$$\sigma_i^2 = \sigma_1^2(1 + \theta)^{i-1}.$$

This alternative unfortunately does not lead to a simplification of the test statistic similar to the above. If we neglect θ^2 and higher powers of θ this alternative and that of Vincent are effectively the same.

In view of these remarks attention is mainly restricted to Vincent's alternative and the corresponding statistic

$$B_v = \frac{\sum_{i=2}^k (i-1)S_i^2}{\sum_{i=1}^k S_i^2}.$$

It should be noted that when the means of the populations are known, the sufficient statistics for the unknown variances are

$$S_1'^2, \dots, S_k'^2 \text{ with}$$

$$S_i'^2 = \sum_{j=1}^{n_i} (X_{ij} - \mu_i)^2$$

and the test statistic reduces to

$$B' = \frac{\sum_{i=2}^k a_i x_{n_i}^2}{\sum_{i=1}^k x_{n_i}^2}$$

2. Vincent's Test

For the case of equal sample sizes Vincent (1961) showed that the uniformly most powerful unbiased critical region of size α for testing H_0 against H_a , with $a_i = (i-1)$, is given by

$$V \geq c(U)$$

where

$$V = \sum_{i=1}^k i S_i^2$$

$$U = \sum_{i=1}^k S_i^2$$

and $c(u)$ is chosen so that under H_0

$$(1) \quad P(V \geq \alpha(U)|U = u) = \alpha.$$

Vincent shows that $\alpha(u)$ is 'approximately' of the form ub with b a constant in which case the critical region is given by large values of $V/U = 1 + B_v$.

The relation $\alpha(u) = ub$ does not hold approximately but exactly. To see this we write equation (1) in the form

$$(2) \quad P(V/U \geq c(U)/U|U = u) = \alpha.$$

It is well known that V/U is stochastically independent of U , and hence a function $c(u)$ satisfying equation (2) is $\alpha(u) = ub$ where

$$P(V/U \geq b) = \alpha \text{ under } H_0.$$

By a direct extension of Vincent's work it can also be shown that for different sample sizes the test, with critical region large values of B_v , is the uniformly most powerful unbiased test for testing H_0 against H_a .

3. Approximation to the Distribution of B_v when H_0 is True

When H_0 is true, B_v can be written as

$$B_v = \sum_{i=2}^k (i-1)Y_i$$

with

$$Y_i = \chi_{N_i}^2 / \sum_{j=1}^k \chi_{N_j}^2, \quad N_i = n_i - 1.$$

Two limiting cases are considered. If k is constant and the $n_i \rightarrow \infty$ then B_v is asymptotically normal, and this also holds when the n_i are constant and $k \rightarrow \infty$.

The derivation of the exact distribution of B_v is cumbersome and we require a suitable small sample approximation.

For $k = 2$, B_v is distributed as a $\beta_1(\frac{N_2}{2}, \frac{N_1}{2})$ variable, and using the fact that $(Y_2, Y_3, \dots, Y_k)$ has a $(k-1)$ -variate Dirichlet distribution, $\sum_{i=2}^k Y_i$ is also distributed as a Beta type one variable. These arguments provide motivation why one should intuitively expect that the distribution of $Q = B_v/(k-1)$ can be approximated by a Beta type one distribution in the case of small n_i and k . Equating the mean and variance of Q , namely

$$E(Q) = \frac{\sum_{i=2}^k N_i(i-1)}{(k-1) \sum_{i=1}^k N_i}$$

$$\text{var}(Q) = \frac{(\sum_{i=1}^k N_i)(\sum_{j=2}^k N_j(j-1)^2) - (\sum_{j=2}^k N_j(j-1))^2}{(k-1)^2 (\sum_{i=1}^k N_i)^2 (\sum_{i=1}^k \frac{N_i}{2} + 1)},$$

with the corresponding parameters of a $\beta_1(p,t)$ variable, the parameters p and t can be selected. In the case of equal sample sizes, namely n , we have

$$p = t = \frac{k+3k(n-1) - 3n + 7}{4(k+1)} - 8.$$

For special small sample cases the approximate and exact exceedance probabilities are compared in Table 1.

TABLE 1:
Comparison of exceedance probabilities

Beta approximation	Exact values		
	$n=3, k=3$	$n=5, k=3$	$n=3, k=4$
0,10	0,1007	0,0992	0,0973
0,05	0,0547	0,0520	0,0515
0,01	0,0140	0,0120	0,0130

For a wide range of intermediate values of k and n the approximate exceedance probabilities are compared with empirical values obtained from 8 000 artificially generated samples (See Table 2). Regarding the approximate values as the true probabilities it is found that all the empirical values lie within two standard deviations from these probabilities.

TABLE 2:
Comparison of exceedance probabilities

Beta approximation	Empirical values		
	$n=3, k=15$	$n=5, k=10$	$n=11, k=5$
0,10	0,0956	0,0998	0,0964
0,05	0,0463	0,0513	0,0464
0,01	0,0086	0,0110	0,0085

These comparisons show that the Beta approximation is quite satisfactory and for equal sample sizes the corresponding approximate percentage points are given in Tables 4, 5 and 6. The ranges of k and n in these tables extend to the point where the Beta distribution can satisfactorily be approximated by the normal distribution.

5. Power Comparison with Bartlett's Test

Since Bartlett's test is also invariant the test based on B_v is uniformly better under Vincent's alternative. This comparison is unfair and to counterbalance we compare the tests with respect to that subclass of H_a for which B_v performs worst relative to Bartlett's test. To keep this comparison mathematically tractable we restrict attention to the case where k is fixed and $n \rightarrow \infty$, and where all sample sizes are equal to n .

Bartlett's test for the homogeneity of variances is based on the statistic

$$M = kn \{ \ln \bar{\sigma}^2 - \sum_{i=1}^k (\ln \hat{\sigma}_i^2)/k \}$$

with

$$\hat{\sigma}_i^2 = S_i^2/n, \quad \bar{\sigma}^2 = \sum_{i=1}^k \hat{\sigma}_i^2/k.$$

By expanding $\ln \hat{\sigma}_i^2$ about $\bar{\sigma}^2$ we obtain

$$M = (n/2) \sum_{i=1}^k (\hat{\sigma}_i^2 - \bar{\sigma}^2)^2 / \bar{\sigma}^4 + R_n$$

$$R_n = -n \sum_{i=1}^k (\hat{\sigma}_i^2 - \bar{\sigma}^2)^3 / 3\Delta_{in}^3$$

where Δ_{in} is bounded by $\bar{\sigma}^2$ and $\hat{\sigma}_i^2$.

Under H_a , with $\theta = d/n^{1/2}$, the statistic $(\frac{n}{2})^{1/2} (\hat{\sigma}_i^2 - \sigma_1^2)/\sigma_1^2$ converges in distribution to a normal variate with mean $a_i d/2^{1/2}$ and unit variance and $\hat{\sigma}_i^2 - \sigma_1^2$ converges in probability to zero. It thus readily follows that Δ_{in}^3 converges in probability to σ_1^2 , R_n converges in probability to zero, and M converges in distribution to a noncentral χ^2 variable with $(k-1)$ degrees of freedom and noncentrality parameter

$$\lambda = d^2 \sum_{i=1}^k (a_i - \bar{a})^2/2.$$

By writing B_v in the form

$$\begin{aligned} B_v &= \sum_{i=1}^k (i-1) \hat{\sigma}_i^2 / k \bar{\sigma}^2 \\ &= \sum_{i=1}^k (i - \frac{k+1}{2}) \hat{\sigma}_i^2 / k \bar{\sigma}^2 + (k-1)/2 \end{aligned}$$

it readily follows that under H_a , with $\theta = d/n^{1/2}$ the limiting distribution of $kn^{1/2} \{B_v - (k-1)/2\} / \{k(k^2-1)/6\}^{1/2}$ is normal with mean $d \sum a_i(i - \frac{k+1}{2}) / \{k(k^2-1)/6\}^{1/2}$ and unit variance.

The subclass of H_a for which B_v is asymptotically least favourable relative to M , is that for which $\sum a_i(i - \frac{k+1}{2})$ is minimized subject to $\sum (a_i - \bar{a})^2$ being constant, say $\sum (a_i - \bar{a})^2 = 1$, and this is achieved when

$$a_1 = \dots = a_{k-1} = 0, a_k = \left(\frac{k}{k-1}\right)^{1/2}.$$

Under this alternative the asymptotic power P_M of M is given by the exceedance probabilities of a noncentral χ^2 distribution with $(k-1)$ degrees of freedom and noncentrality parameter $d^2/2$, while the asymptotic power P_B of B_v is given by the exceedance probabilities of a normal distribution with mean $d \{3/2(k+1)\}^{1/2}$ and unit variance. Since P_M is symmetric in d and differentiable at $d=0$

$$\left. \frac{\partial P_M}{\partial d} \right|_{d=0} = 0$$

$$\text{while } \left. \frac{\partial P_B}{\partial d} \right|_{d=0} \text{ is proportionate to } \{3/2(k+1)\}^{1/2}.$$

For d sufficiently small P_B will thus exceed P_M . On the other hand for larger d inspection shows that P_M exceeds P_B and for increasing k this effect becomes more marked.

TABLE 3:
Comparison of P_M and P_B for $k = 5; 10; 20$ at the 0.05 level of significance

k = 5			k = 10			k = 20		
λ	P_M	P_B	λ	P_M	P_B	λ	P_M	P_B
0.91	0.1	0.17	1.40	0.1	0.15	2.09	0.1	0.14
2.40	0.2	0.29	3.53	0.2	0.25	5.05	0.2	0.21
3.74	0.3	0.39	5.33	0.3	0.33	7.47	0.3	0.27
5.05	0.4	0.48	7.05	0.4	0.40	9.73	0.4	0.32
6.42	0.5	0.56	8.81	0.5	0.46	12.00	0.5	0.37
7.92	0.6	0.63	10.71	0.6	0.53	14.41	0.6	0.42
9.68	0.7	0.71	12.89	0.7	0.59	17.14	0.7	0.47
11.94	0.8	0.79	15.65	0.8	0.66	20.56	0.8	0.53
15.41	0.9	0.87	19.83	0.9	0.75	25.65	0.9	0.61

These power comparisons, although fragmentary, suggest that at its worst the test based on B_V is comparable with M within the class of alternatives H_a .

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TABLE 6 : Approximate upper 1% critical values for $Q = B_V/(k-1)$

$k \backslash n$	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
2	0,999	0,990	0,967	0,941	0,917	0,894	0,875	0,858	0,843	0,829	0,817	0,806	0,796	0,787	0,779	0,771	0,764	0,758
3	0,974	0,917	0,871	0,836	0,809	0,787	0,770	0,755	0,742	0,731	0,722	0,713	0,706	0,699	0,693	0,687	0,682	0,677
4	0,931	0,858	0,810	0,777	0,753	0,734	0,719	0,706	0,695	0,686	0,678	0,671	0,665	0,659	0,654	0,649	0,645	0,641
5	0,894	0,817	0,771	0,741	0,719	0,702	0,688	0,677	0,667	0,659	0,652	0,646	0,641	0,636	0,632	0,628		
6	0,865	0,787	0,744	0,716	0,695	0,680	0,667	0,657	0,649	0,641	0,635	0,630	0,625	0,620				
7	0,841	0,764	0,723	0,697	0,678	0,663	0,652	0,643	0,635	0,628	0,623							
8	0,821	0,746	0,707	0,682	0,664	0,651	0,640	0,632	0,624									
9	0,804	0,731	0,694	0,670	0,653	0,641	0,631	0,623										
10	0,790	0,719	0,683	0,660	0,644	0,632	0,623											
11	0,777	0,708	0,673	0,652	0,637	0,625												
12	0,766	0,699	0,665	0,645	0,630													
13	0,756	0,691	0,658	0,638	0,625													
14	0,747	0,683	0,652	0,633														
15	0,739	0,677	0,647	0,628														
16	0,732	0,672	0,642															
17	0,725	0,666	0,637															
18	0,719	0,661	0,633															
19	0,714	0,657	0,629															
20	0,708	0,652																
21	0,704	0,649																
22	0,699	0,645																
23	0,695	0,642																
24	0,691	0,639																
25	0,687	0,636																
26	0,684	0,633																
27	0,680																	
28	0,677																	
29	0,674																	
30	0,671																	
32	0,666																	
34	0,661																	
36	0,657																	
40	0,649																	

$k \backslash n$	20	21	22	23	24	25	26	27	28	29	30	32	34	36	38	40	50
2	0,752	0,746	0,741	0,736	0,731	0,727	0,723	0,719	0,715	0,711	0,708	0,702	0,696	0,691	0,686	0,681	0,662
3	0,672	0,668	0,664	0,661	0,658	0,654	0,651	0,649	0,646	0,644	0,641	0,637	0,633	0,629			
4	0,638	0,634	0,631	0,628	0,626	0,623	0,621	0,618									